



SOLAR ENERGY PRODUCTION ARCHITECTURE

Enterprise Systems Architecture

ABSTRACT

In this document we will outline how we created a layered reference architecture for SolarSphere, a multinational renewable energy company.

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TABLE OF CONTENTS

Introduction	3
Architecture Vision.....	3
Business Context.....	3
Goals	3
Scope	3
Guiding Principles	3

INTRODUCTION

ARCHITECTURE VISION

BUSINESS CONTEXT

SolarSphere Energy is a large multinational renewable energy company. Their focus is on operating and managing large-scale solar photovoltaic power plants. They have operations across North America, Europe, the Middle East, Africa, and Asia-Pacific.

SolarSphere owns and manages:

- Utility-scale solar farms
- Distributed solar generation facilities
- Battery Energy Storage Systems (BESS)
- Grid interconnection infrastructure
- Regional Operations Control Centers
- Asset maintenance and field service operations

The company generates electricity for utility companies, commercial customers, energy markets, and power purchase agreement (PPA) clients.

GOALS

SolarSphere requires a unified enterprise architecture to support:

- Real-time monitoring of solar generation assets and performance
- Solar plant operations and energy production optimization
- Grid integration and energy dispatch management
- Predictive maintenance and asset lifecycle management
- Weather forecasting and generation forecasting
- Battery energy storage monitoring and optimization
- Environmental, sustainability, and ESG reporting
- Integration with ERP, SCADA, Asset Management, Energy Trading, and Grid Management systems

SCOPE

Our Information Architecture will outline how data such as machine efficiency, weather reports, energy generation and sustainability reports should flow through the company.

Our application architecture will show how the databases, data-collection software and prediction software integrate with the existing management systems.

GUIDING PRINCIPLES

SolarSphere needs to maximize energy collection, customer satisfaction and profits, to do this it is important that they have a clear way of structuring data and technology within the company to avoid information silo.